

# ANDREW Y. KANG

<https://andrew-kang.github.io>

[aykang@umich.edu](mailto:aykang@umich.edu)

RESEARCH Formal methods, cyber-physical systems, control theory, category theory

## EDUCATION **University of Michigan – Ann Arbor**

Ph.D., Electrical and Computer Engineering, 2026 – 2031. (expected)

- Advisor: Prof. Inigo Incer

M.S., Electrical and Computer Engineering, 2026 – 2028. (expected)

## **Cornell University**

B.A., Mathematics and Computer Science, 2023 – 2026. (graduated 1-year early)

- GPA: 4.0/4.0 (4.13/4.3 on Cornell scale)
- Magna Cum Laude in Mathematics with Distinction in all Subjects
- Phi Beta Kappa ( $\Phi\text{BK}$ ), Phi Kappa Phi ( $\Phi\text{K}\Phi$ )

## EXPERIENCE **University of Michigan – Ann Arbor**

Ann Arbor, MI

*Graduate Student*

August 2026 – present

- Completed coursework and conducted research projects.

## **Cornell University**

Ithaca, NY

*Project Lead & Undergraduate Researcher*

February 2024 – May 2026

Advisor: Prof. Sainyam Galhotra

- Studied existing literature in programming languages and databases.
- Designed syntax and semantics for TQL, a novel domain-specific language for data discovery, leveraging techniques from programming languages and databases research.
- Implemented a proof-of-concept system for language-driven data discovery in Python.
- Drafted two manuscripts compiling research results.

*Head Teaching Assistant*

August 2024 – May 2026

Courses: Functional Programming, Formal Verification, Machine Learning

- Supervised office hour assignments of 45+ TAs.
- Oversaw grading sessions for assignments & exams.
- Led proctoring team for course exams.
- Led recitation sessions of 40+ students.

## **Computational & Quantitative Social Science at Cornell**

Ithaca, NY

*Student Club Founder & President*

October 2023 – December 2025

- Founded and led a student organization interested in building data science and data engineering tools for research in the social sciences.

## AWARDS & PRESS **Featured Graduate for Cornell CIS Class of 2026**

*Cornell Bowers College of Computing and Information Science (CIS)*

May 2026

- Selected by Cornell CIS to be the sole representative for the CS graduates of the Class of 2026.
- Interviewed and featured across official university websites and social media channels (Linkedin, Instagram, X and Facebook). [Andrew Kang ‘26: Making the most of undergraduate research opportunities.](#)

**2025-2026 CRA Outstanding Undergraduate Researcher Award (Honorable Mention)**  
*Computing Research Association (CRA)* December 2025

- Nominated by Cornell Math Department; received honorable mention from Computing Research Association (CRA) amongst undergraduates across North America in recognition of outstanding potential in computing research.

**Cornell CS Department Course Staff Award (Nominee)**  
*Cornell Bowers College of Computing and Information Science (CIS)* December 2024

- Nominated by faculty for course staff award in recognition of excellence in teaching.

**Summer Research Experience for Undergraduates (Grant)**  
*Cornell Bowers College of Computing and Information Science (CIS)* June 2024

- Selected and received \$7000 grant to participate in Cornell's BURE summer REU.
- Selected and received additional \$1000+ travel grant for conference presentation through Cornell's BURE extension.

PUBLICATIONS      **Andrew Y. Kang**, Yashnil Saha, and Sainyam Galhotra. **Towards General-Purpose Data Discovery: A Programming Languages Approach**. arXiv Preprint, 2025.

**Andrew Y. Kang**, and Sainyam Galhotra. **TQL: Towards Type-Driven Data Discovery**. IEEE International Conference on Big Data (**BigData**), 2024.

PRESENTATIONS      **Cornell Entrepreneur of the Year: Undergraduate Research Talk** April 2025

- Represented undergraduate REU research for the college of computing and information science.
- Orally presented findings to alumni 'Entrepreneur of the Year' award recipient, John Bicket '02.

**IEEE BigData 2024: Conference Paper Presentation (Oral)** December 2024

- Orally presented the paper, *TQL: Towards Type-Driven Data Discovery*.

**Cornell CIS REU: BURE Symposium (Poster)** August 2024

- Presented poster summarizing results of summer REU research.

PROJECTS      **Eta Compiler (Cornell CS 4120/4121 - Compilers)** January 2026 - May 2026

- Built a complete optimizing compiler targeting x86-64 for Eta, a C-like programming language.
- Implemented lexer, parser, typechecker and intermediate and assembly code generation.
- Implemented dataflow analysis and optimization passes (including constant folding, copy propagation, dead-code elimination, and local-value numbering).

|            |                                        |                                     |                               |
|------------|----------------------------------------|-------------------------------------|-------------------------------|
| COURSEWORK | • Advanced Linear Algebra <sup>†</sup> | • Real Analysis I & II <sup>†</sup> | • Compilers (incl. Practicum) |
|            | • Abstract Algebra <sup>†</sup>        | • Numerical Analysis                | • Systems Programming         |
|            | • Kleene Algebra*                      | • Probability Theory                | • Functional Programming      |
|            | • Category Theory*                     | • Analysis of Algorithms            | • Data Structures & OOP       |
|            | • Applied Logic                        | • Machine Learning                  | • Computer Organization       |
|            |                                        |                                     |                               |

<sup>†</sup>Indicates honors-level coursework

\*Indicates graduate-level coursework

RELEVANT SKILLS      Formal Languages: C/C++, Python, Java, Assembly, Rust, OCaml, Rocq, Lean, SQL,  $\text{\LaTeX}$   
Natural Languages: English (native); French, Mandarin (proficient); Latin (intermediate)